QP CODE: 25900140

MAHATMA GANDHI UNIVERSITY

MGU-UGP (HONOURS} REGULAR EXAMINATIONS MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) – MG2DSCMBG100 – PERSPECTIVES OF MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum marks: 50**

PART A

Short Answer type questions, Answer any 10

Each question carries 2 marks

1. **Which type bacteria lacks a cell wall.** **[K] / [CO1]**
   Mycoplasma (2)

2. **Explain types of hyphae.** **[U] / [CO1]**
   Explain Septate (1) and Aseptate hyphae (1)

3. **Name the method for estimating the size and shape of viruses.** **[K] / [CO1]**
   Electron microscopy (2)

4. **What is a helical virus?** **[K] / [CO1]**
   Define helical symmetry with any one example (2)

5. **Define culture media.** **[K] / [CO2]**
   Define culture media with one example (2)

6. **Define alkaliphiles with example.** **[K] / [CO2]**
   Define Alkaliphiles (1) Give one example (1)

7. **Define heterotrophic bacteria.** **[K] / [CO2]**
   Define heterotrophs with one example (1)

8. **Define basal media with suitable example.** **[K] / [CO2]**
   Define basal media (1) one example (1)

9. **Name two anaerobic culture media.** **[K] / [CO1]**
   Write the names of any two anaerobic media (1 mark each)

10. **What type of sample is best suited for direct microscopic method.** **[K] / [CO3]**
    Accurate in low concentration sample (2).

11. **Illustrate the mode of reproduction in Nocardia.** **[U] / [CO3]**

Explain Fragmentation and formation conidia (2).

12. List the characteristics of stationary phase. [K] / [CO3]

Growth rate slows and becomes equal to death rate, the no. of new cells is balanced by the no. of cells dying, high cell density as a result nutrients become limited and waste products accumulate, formation of endospores, release of toxins & stress responses, write any two major points (2).

PART B

Short Essay questions, Answer any 4

Each question carries 5 marks.

13. Discuss the role of water in microbial growth. [U] / [CO2]

Explain role of water – In absorption, metabolic reactions, protection, chemical reactant (2)

Water activity and its effect on microbial growth (3).

14. Explain the nutritional classification of bacteria and explain each with examples. [U] / [CO2]

Explain Phototrophs and Chemotrophs with examples (2), Autotrophs and Heterotrophs with examples (2), Lithotrophs and Organotrophs with examples (1)

15. Compare and contrast the pour plate, spread plate and membrane filtration methods for bacterial enumeration. [U] / [CO3]

Explain the Principle, method - Pour plate technique (1) Spread plate technique (1) Membrane filtration method (1). Explain the major differences (2).

16. What is CFU? How do you calculate CFU in SPC? [K] / [CO3]

Colony Forming Unit (1), Explain colony and CFU (1). Explain Dilution and Dilution factor (1), Formula & Calculation of viable count (2).

17. Explain with diagrams the different types of flagellar arrangement in bacteria. [U] / [CO1]

Explain – Types of flagellar arrangement (1), Diagram - Monotrichous, Lophotrichous, Amphitrichous and Peritrichous (4)

18. Explain budding in yeast. [U] / [CO1]

Explain stages of budding in yeast (3). Diagram (2).

Part C

Long Essay Question. Answer any 1 question
Each question carries 10 marks

19. Explain the principle and working of turbidimetric method for bacterial enumeration. [U] / [CO3]

Explain Principle of Turbidometry – Determination of cell mass – indirectly using spectrophotometry (4), Instrumentation and method (4), Calculation - Percentage of transmittance of culture dilutions is measured and converted to OD - formula (2)

20. Explain the classification of microorganisms based on oxygen requirement with suitable examples. [U] / [CO2]

Explain with examples - Obligate Aerobes (2), Facultative anaerobes (2), Obligate anaerobes (2), Microaerophiles (2), and Aerotolerant anaerobes (2).